

Quick Check

Lesson 7: Putting It All Together: Final Explanation of How Animals Keep a Balanced Internal Environment and Remove Toxic Wastes

9 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.3: Excretion and Homeostasis in Animals | Lesson 7

A goat in Kajiado is kept without water for a whole day on a hot afternoon. What is the most likely result?

- A** Its urine becomes more dilute because the kidney stops working to save energy
- B** Its urine stays the same because goats only control water by drinking less
- C** Its blood water level rises because the body makes extra water from urine
- D** Its urine becomes more concentrated because ADH makes the kidney reabsorb more water

Learners cut open a goat kidney and review how a nephron works. Which statement is correct?

- A** The loop of Henle collects urine and carries it to the bladder
- B** The glomerulus reabsorbs water and the tubule filters the blood
- C** The pelvis filters the blood and the cortex stores the urine
- D** The glomerulus filters the blood and the tubule reabsorbs useful substances such as water

Which row correctly matches the animal to its main nitrogenous waste?

- A** Freshwater fish: uric acid; mammals: ammonia; insects: urea
- B** Freshwater fish: ammonia; mammals: uric acid; insects and birds: urea
- C** Freshwater fish: ammonia; mammals: urea; insects and birds: uric acid
- D** Freshwater fish: urea; mammals: uric acid; birds: ammonia

A tilapia from Lake Victoria produces large amounts of dilute urine. What is the best explanation?

- A** It makes dilute urine because it does not need to excrete nitrogenous wastes
- B** Water enters its body by osmosis, so its kidneys remove the excess water as dilute urine
- C** Its body is saltier than the lake water, so it loses water by osmosis and must make dilute urine
- D** It drinks a lot of lake water and must remove the extra salt in urine

On a hot afternoon a dog pants with its tongue hanging out, while a lizard sits still on a warm rock. Which statement is correct?

- A** Panting and basking both make the animals produce more urine to lose heat
- B** Panting warms the dog's body, and the lizard stays still because it makes its own body heat
- C** Panting mainly removes nitrogenous waste from the dog, and the lizard cannot control its temperature
- D** Panting increases evaporation to cool the dog, and the lizard controls its temperature by behaviour such as basking or moving to shade

After a meal rich in sugar, a person's blood glucose rises above the set point. Which sequence shows negative feedback?

- A** Insulin is released, blood glucose rises further, and the pancreas stops working
- B** Glucagon is released, the liver releases more glucose, and blood glucose rises even further
- C** Insulin is released, cells and the liver take up glucose, and blood glucose falls back towards the set point
- D** The kidney makes ADH, and blood glucose returns to the set point

A patient passes very large volumes of dilute urine because the body does not make or respond to enough ADH. What explains this?

- A** The liver produces too much urea, which pulls water into the urine
- B** The skin reabsorbs water and stops sweating
- C** The glomerulus filters too little blood
- D** The kidney tubules reabsorb too little water, so more water is lost in urine

When learners add the liver to their final model to show where wastes come from, which statement is correct?

- A** The liver makes ADH, which turns ammonia into uric acid
- B** The liver filters blood in nephrons and forms urine
- C** The liver makes sweat, which carries urea out through the skin only
- D** The liver removes the amino group from excess amino acids (deamination) and forms urea, which the kidneys excrete

Which statement best explains why excretion and homeostasis are important for the survival of body cells?

- A** Excretion adds water to cells, and homeostasis makes the external environment stable
- B** Excretion and homeostasis are only needed when an animal is ill or in a hot place
- C** Excretion keeps the body temperature steady, and homeostasis removes urea from the blood
- D** Excretion removes toxic metabolic wastes from cells, and homeostasis keeps conditions such as temperature and water level steady

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).